

# A Parent-Aware Computer-Assisted Lower Bound for Packing 17 Unit Squares in a Square

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<https://github.com/Mira-acc/17squares>

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## Abstract

Let  $s(n)$  denote the least side length of a square that contains  $n$  pairwise interior-disjoint unit squares, with arbitrary orientations. We prove by an exact computer-assisted base certificate together with a finite analytic transport argument that

$$s(17) > \frac{46129999999859}{9997499999900} = 4.614153538416645696808\dots$$

The base certificate places a nonnegative rational measure in a square of side  $L = 4.613$  and excludes seventeen squares of side  $A = 3999/4000$ . It is *parent aware*: for each continuous parent-angle interval it selects a closed core strictly inside the parent and checks that core only over the complete center domain attainable by legal parents in that interval. The measure has 3,280 atoms in 417  $D_4$ -orbits, total mass 16.999991644, and every selected core has mass at least 1.000000030. A recentering theorem then reuses these already certified core-center domains for slightly smaller parents and a slightly smaller container, yielding the displayed stronger rational bound without a new translation sweep. Finally, we generalize the charging language from point measures and threshold atoms to *intersection-trigger* charges, which compile finite overlap incompatibilities into budgeted spatial resources while retaining a two-dimensional event representation.

## 1 Problem and current interval

For  $L > 0$  write  $Q_L = [0, L]^2$ . Let  $s(n)$  be the minimum  $L$  for which  $n$  pairwise interior-disjoint unit squares can be placed in  $Q_L$ . The minimum is attained by compactness. The best known 17-square construction is due to John Bidwell and has side

$$L_B = 4.6755300936045509516\dots$$

according to the high-precision catalogue maintained by Ellsworth and the historical survey literature[1, 4].

Our main result is the following.

**Theorem 1.** *No packing of seventeen unit squares exists in a square of side*

$$S = \frac{46129999999859}{9997499999900}.$$

*Consequently*

$$s(17) > \frac{46129999999859}{9997499999900} = 4.614153538416645696808\dots$$

Thus the currently certified interval retained by this repository is

$$4.614153538416645696808 \dots < s(17) \leq 4.6755300936045509516 \dots$$

The lower-bound proof is independent of any assumption that a near-optimal packing resembles Bidwell’s construction.

## 2 How the lower bound developed

The current certificate is best understood as the end of a sequence of increasingly flexible transversal arguments. Table 1 lists results that directly influenced this project or provide important nearby benchmarks. Some were subsequently superseded; inclusion in the table does not assert coauthorship of the present theorem.

The classical unavoidable-point method is the integral special case of the weighted method used below. If every unit square contains one of sixteen points strictly in its interior, then seventeen disjoint interiors would need seventeen distinct points. The August 2026 certificates of Mira and Fort made that argument completely finite by subdividing the three-dimensional pose space and checking each leaf with exact integer arithmetic[6, 5].

Burns’ proposal made the decisive relaxation: assign nonnegative weights to many points, require every possible square to capture a prescribed amount of mass, and keep the total mass below 17[7]. Massaccesi used linear programming to improve the measure, and Levy turned the architecture into a reusable exact event-cell framework, obtaining 4.59[11]. The present project’s subsequent 4.607 and 4.613 certificates used the same fractional principle with more refined support and angular handling.

Guzhou0806’s R012 certificate identified a further overconstraint. An auxiliary core may fit close to the wall even though the larger parent square from which it is supposed to arise cannot fit there. R012 therefore checks a core only over the complete center domain attainable by parents in the corresponding angle interval and allows the core direction and size to vary with that interval[12]. The theorem below formalizes that reduction and is the proof interface used by our final certificate.

For source links, verification status, and a more detailed division between direct dependencies and parallel historical work, see the repository file `CONTRIBUTORS.md`.

### 2.1 Stage I: unavoidable points and exact pose-space subdivision

Green’s classical argument places sixteen witnesses so that every admissible unit square meets at least one of them. The proof is then a pigeonhole argument: seventeen disjoint interiors cannot be assigned seventeen distinct witnesses from a set of size sixteen. The weakness is geometric rather than combinatorial—one must prove a universal statement over all translations and orientations of one square.

The 2026 exact-point certificates, including Brandwijk’s 4.45, Mira’s 4.450837 and later 4.468292, and Fort’s 4.456575, attack precisely this universal geometric step. They partition the three-dimensional pose space (center  $x$ , center  $y$ , orientation) into finitely many rational boxes and certify each box either infeasible or pierced by a witness point. Fort’s work is especially useful historically because it independently optimized the same basic exact architecture. These results differ numerically but share the same integral counting invariant:

$$\text{total witness budget} = 16 < 17.$$

Table 1: Selected lower-bound lineage for  $n = 17$ . Ordered by bound, not date; decimals are for orientation. Dates refer to source publication or retained records; Green’s date is the reported communication year.

| Date                      | Contributor / source       | Bound                                        | Main contribution                                                                                                                                                                                           |
|---------------------------|----------------------------|----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2000                      | Trevor Green, via Friedman | 4.445208...                                  | Classical sixteen-point unavoidable-set scaffold; the long-standing published benchmark.                                                                                                                    |
| 2026-07-18                | Kim Brandwijk              | 4.45                                         | Exact sixteen-point certificate in the 2026 public certificate sequence[8].                                                                                                                                 |
| 2026-08-10;<br>2026-08-11 | Mira                       | 4.450837, later<br>4.468292                  | Exact pose-space subdivision for sixteen unavoidable rational points; the later version adds triangle-piercing leaves.                                                                                      |
| 2026-08-11                | Stanislav Fort             | 4.456575                                     | Independent optimization of the same exact sixteen-point subdivision architecture.                                                                                                                          |
| 2026-08-06                | Sam Burns                  | 4.4811                                       | Pure weighted atomic certificate: replace sixteen unit witnesses by a fractional covering measure.                                                                                                          |
| 2026-08-21                | Gustavo Massaccesi         | 4.5058                                       | LP-derived weighted measure with a smaller support, checked by an exact direction/event sweep[9].                                                                                                           |
| 2026-08-16                | anabologyco-maker          | 4.5705                                       | Weighted measure with an exact orientation partition and a Lean layer for the finite computation.                                                                                                           |
| 2026-09-04                | Joshua Levy                | 4.59                                         | Reusable weighted-certificate generator, exact event-cell verification, independently checked interval coverage, and later dilation/parent-center machinery.                                                |
| 2026-09-07                | Mira continuation          | 4.607028598640..., then<br>4.613028635886... | Finer angular control, mixed support geometry, dual-guided column generation, and exact dilation.                                                                                                           |
| 2026-09-19                | Guzhou0806 / N17           | 4.613046130461...                            | Parent-angle catalogue: check only centers attainable by legal parents and allow the interior core to depend on the parent-angle interval.                                                                  |
| 2026-09-19                | This work                  | 4.614153538416...                            | Adaptive parent-aware intervals plus reweighted/wall-adjusted support give 4.614153538384...; an analytic recentering transfer then gives the displayed stronger rational bound without a new global sweep. |

## 2.2 Stage II: fractionalize the witness budget

Burns’ weighted formulation replaces the discrete set of sixteen unit witnesses by a nonnegative atomic measure of total mass below 17. Every admissible square must capture at least unit mass. The pigeonhole principle is replaced by additivity over disjoint interiors:

$$17 \leq \sum_{i=1}^{17} \mu(U_i) \leq \mu(Q_L) < 17.$$

This relaxation can place small amounts of mass at many locations and is strictly more flexible than sixteen points of weight one.

Massaccesi showed that the support and weights could be chosen effectively by linear programming, raising the numerical bound substantially. The anabologyco-maker certificate pushed the same weighted idea much farther with a large exact orientation partition and an additional Lean-checked finite layer. Levy then supplied a reusable generator and exact event-cell sweep, together with an interval-based cross-check and later dilation machinery. In our subsequent continuation, finer angular control, mixed wall/interior support, and dual-guided column generation raised the weighted method through 4.607028598640... and then 4.613028635886....

Mathematically, all of these results retain the same global invariant:

$$\text{least charge per selected core} \times 17 > \text{total available budget}.$$

The improvements come from making the finite coverage problem easier to satisfy without increasing that budget.

### 2.3 Stage III: make the core transport parent aware

The remaining inefficiency is subtler. Earlier weighted verifiers asked for coverage of every auxiliary core position allowed by the core’s own geometry. But a core is not a physical packed square; it is only a device chosen inside a larger parent. Some core centers near the container wall cannot arise from any legal parent at the corresponding orientation.

Guzhou0806’s R012 exploits this distinction. It partitions the parent-angle continuum into intervals, chooses one safe core recipe per interval, and checks coverage only over the union of centers attainable by the actual parents in that interval. This is a weakening of the verification obligation while preserving the physical theorem. The present work keeps that relaxation, makes the interval/core selection adaptive, and then changes the underlying measure once the frozen R012 measure reaches an exact obstruction.

This gives a precise sense in which the current proof *subsumes* the earlier methods: the final counting theorem still contains the classical pigeonhole invariant, fractional weights contain the exact-point method as an integral special case, and parent-aware transport removes a constraint that is irrelevant to the physical packing problem.

## 3 Parent-aware weighted certificates

The counting argument has a slightly more general form that separates the geometric question from the particular choice of atoms.

**Lemma 2** (Budgeted disjoint-core principle). *Let  $C(K) \geq 0$  be a charge assigned to closed cores  $K \subseteq Q_L$ . Suppose there is a global budget  $\mathcal{B}$  such that every finite pairwise-disjoint family of cores satisfies*

$$\sum_i C(K_i) \leq \mathcal{B}.$$

*If every legal parent square admits a closed core strictly inside it with charge at least  $\gamma$ , and if*

$$\mathcal{B} < n\gamma,$$

*then  $n$  pairwise interior-disjoint parents cannot fit in the container.*

*Proof.* Choose one certified core from each hypothetical parent. Since every core is a closed subset of its parent interior, the selected cores are pairwise disjoint. Their total charge is at least  $n\gamma$ , contradicting the budget.  $\square$

Ordinary point measures are the simplest realization: if  $\mu = \sum_j w_j \delta_{p_j}$ , set  $C(K) = \mu(K)$  and  $\mathcal{B} = \mu(Q_L)$ . The same lemma also contains Levy's threshold atoms. For a finite set  $S$ , integer  $k \geq 1$ , and weight  $w \geq 0$ , a threshold atom  $(S, k, w)$  contributes  $w$  to  $C(K)$  when  $|K \cap S| \geq k$ . For pairwise-disjoint cores, the traces  $K_i \cap S$  are disjoint, so at most  $\lfloor |S|/k \rfloor$  cores can activate the atom. Its global budget is therefore

$$w \left\lfloor \frac{|S|}{k} \right\rfloor.$$

Budgets add over independent point and threshold atoms. The current 4.614153... certificate uses point atoms only; this more general statement is included to make precise how the parent-aware transport composes with the stronger charging rules already present in the literature.

We now specialize to a finite nonnegative measure on  $Q_L$ . Fix an outer side  $L$  and a parent side  $A$ , and let

$$\mu = \sum_j w_j \delta_{p_j}.$$

We will associate to every legal parent square  $P \subseteq Q_L$  of side  $A$  a closed core  $K(P)$  satisfying

$$K(P) \subset \text{int } P.$$

Different parent orientations may use different core sides and directions.

**Proposition 3** (Parent-aware counting principle). *Suppose every legal side- $A$  parent square  $P \subseteq Q_L$  admits a selected closed core  $K(P) \subset \text{int } P$  with*

$$\mu(K(P)) \geq \gamma.$$

*If*

$$17\gamma > \mu(Q_L),$$

*then seventeen pairwise interior-disjoint side- $A$  squares do not fit in  $Q_L$ .*

*Proof.* If  $P_1, \dots, P_{17}$  had pairwise disjoint interiors, then their selected closed cores would be pairwise disjoint sets. Nonnegativity gives

$$17\gamma \leq \sum_{i=1}^{17} \mu(K(P_i)) = \mu\left(\bigcup_{i=1}^{17} K(P_i)\right) \leq \mu(Q_L),$$

a contradiction. □

The usual weighted proof is the special case  $A = 1$ ,  $\gamma = 1$ , with one uniform family of cores. Proposition 3 shows that neither normalization is necessary: only the ratio between the least selected-core mass and the total mass matters.

### 3.1 A finite certificate theorem that subsumes the recent methods

The computational proofs discussed in Section 2 differ mostly in how they establish the hypothesis of Proposition 3. It is useful to isolate a single theorem that contains the current parent-aware architecture and makes those differences explicit.

Let  $G = D_4$  be the symmetry group of the outer square. Fold the orientation of an individual physical square into the interval  $[0, \pi/4]$  using its own quarter-turn symmetry and, when needed, a reflection of the container. Let  $u = \tan(\phi/2)$  be the folded half-angle parameter.

**Theorem 4** (Finite parent-aware weighted certificate). *Fix  $n \geq 2$ , outer side  $L$ , parent side  $A$ , a finite nonnegative  $G$ -invariant measure  $\mu$  on  $Q_L$ , and  $\gamma > 0$ . Suppose a finite collection of closed half-angle intervals*

$$I_j = [a_j, b_j]$$

*covers the full folded orientation range. For each  $I_j$  choose a rational core half-angle  $t_j$  and core side  $B_j > 0$ . Assume:*

- (C1) the intervals meet without gaps and cover every folded physical orientation;*
- (C2) for every  $u \in I_j$ , the concentric side- $B_j$  square at direction  $t_j$  lies strictly inside the side- $A$  parent at direction  $u$ ;*
- (C3)  $D_j$  is a closed set containing every legal center of every side- $A$  parent whose folded half-angle lies in  $I_j$ ;*
- (C4) every side- $B_j$  core at direction  $t_j$  with center in  $D_j$  has  $\mu$ -mass at least  $\gamma$ ;*
- (C5)  $\mu(Q_L) < n\gamma$ .*

*Then no packing of  $n$  side- $A$  squares exists in  $Q_L$ . Consequently,*

$$s(n) > \frac{L}{A}.$$

*Proof.* Take a hypothetical packing and fold each parent separately. Condition (C1) places its orientation in at least one  $I_j$ . By (C2) choose the corresponding closed core strictly inside that parent; by (C3) its center belongs to the domain on which (C4) was verified.  $G$ -invariance lets us unfold without changing the core's charge. The selected cores lie in pairwise disjoint parent interiors, hence are disjoint sets. Summing their charges gives

$$n\gamma \leq \sum_i \mu(K_i) \leq \mu(Q_L),$$

contradicting (C5). Scaling a unit-square packing at side  $L/A$  by  $A$  would produce the forbidden parent packing. Finally, compactness of the feasible packing space upgrades exclusion at the endpoint to the strict inequality.  $\square$

This theorem separates three logically different tasks: *geometry of transport* (C1–C3), *finite coverage* (C4), and *global counting* (C5). The sequence of recent improvements can be read as successively weakening unnecessary requirements in those three tasks.

**Sixteen unavoidable points.** Take integer atom weights, total mass  $n - 1$ , and prove directly that every physical parent interior captures an atom. This is the integral predecessor of (C4)–(C5). Mira's and Fort's exact pose-space trees differ from the present proof mainly in how the universal coverage statement is finitized.

**Uniform weighted cores.** Burns, Massaccesi, Levy, and our earlier weighted certificates allow fractional weights but use a common core rule over a finite orientation net. The strict-containment lemma transports each arbitrary parent to a nearby checked core. This is Theorem 4 with  $D_j$  taken to be the full domain where the auxiliary core itself fits. That domain is safe but can be larger than the set of centers realizable by physical parents.

**Parent-aware cores.** Guzhou's R012 observes that (C3) only needs a superset of *legal parent centers*, not every location where an auxiliary core happens to fit. It also lets  $(t_j, B_j)$  vary from interval to interval. The present theorem retains that relaxation and adds adaptive interval refinement plus a new measure.

Thus the current proof does not merely place the earlier record values in a timeline: it uses a theorem whose hypotheses expose exactly which constraints each successive method removed.

## 4 Continuous parent angles and exact center domains

Represent a folded parent orientation by the half-angle parameter

$$u = \tan(\phi/2), \quad 0 \leq u < 1.$$

Then

$$c(u) = \frac{1 - u^2}{1 + u^2}, \quad s(u) = \frac{2u}{1 + u^2},$$

and the axis-parallel width factor of a square at that orientation is

$$f(u) = c(u) + s(u) = \frac{1 + 2u - u^2}{1 + u^2}.$$

A side- $A$  parent with parameter  $u$  can have its center precisely in

$$\left[ \frac{Af(u)}{2}, L - \frac{Af(u)}{2} \right]^2.$$

**Lemma 5** (Complete parent-center envelope). *For a closed parent-angle interval  $[a, b] \subset [0, 1)$ , the union of all legal parent-center domains is the square*

$$D_{a,b} = [r, L - r]^2, \quad r = \frac{A}{2} \min\{f(a), f(b)\}.$$

*Proof.* A direct differentiation gives

$$f'(u) = \frac{2(1 - 2u - u^2)}{(1 + u^2)^2}.$$

Hence  $f$  has one interior maximum and no interior minimum. The legal center squares are nested as their half-width changes, so their union is determined by the minimum value of  $f$ , attained at an endpoint.  $\square$

For a candidate core with half-angle parameter  $t$  and side  $B$ , put  $(c_t, s_t) = (c(t), s(t))$ . For a parent parameter  $u$ , its width across either parent normal is bounded by

$$B(c_t c(u) + s_t s(u) + |s_t c(u) - c_t s(u)|).$$

On every retained interval the verifier checks that the relative angle stays within  $\pi/4$ , so this factor is monotone with the absolute relative angle and its maximum occurs at an interval endpoint. The strict rational inequality

$$B \max_{u \in \{a,b\}} (c_t c(u) + s_t s(u) + |s_t c(u) - c_t s(u)|) < A \tag{1}$$

therefore proves that the concentric closed core lies strictly inside every parent throughout the whole continuous interval.

The parent-center envelope matters quantitatively. In the audited R012 data, one fixed auxiliary core has minimum captured mass 0.985238 if every center where the core itself fits is demanded, but minimum 1.000288 on the complete legal-parent envelope. The deficient all-core placement is unattainable by a legal parent. This exact ablation is retained with the current certificate.

## 5 Exact translation sweep

For one catalogue entry  $(a, b, t, B)$ , Lemma 5 supplies the exact closed center square  $D_{a,b}$ . Rotate center coordinates into the selected core frame. Each atom then contributes its weight exactly on a closed axis-parallel square of side  $B$  in center space.

The C++ verifier sweeps all vertical rectangle events and uses clamped extrema of the rotated parent-center domain to determine each slab's reachable vertical projection. Within an open slab, the active rectangles are fixed. Horizontal events partition the reachable vertical projection into open cells of constant charge. Every positive-area reachable cell is checked, and a rational point in every reported minimum cell is directly recounted. All direction vectors are represented by rational half-angle parameters, and all geometry is reduced to integer or rational comparisons.

Closed boundaries cannot conceal a lower charge. Every boundary placement is a limit of non-event interior placements. Because only finitely many cells exist, a subsequence approaches through one fixed cell; closed capture preserves all atoms present in that cell at the limit, and nonnegative weights prevent a decrease. Thus the finite sweep proves the minimum over the entire closed parent-center domain.

The certificate uses the eight symmetries of the outer square to store atoms in  $D_4$ -orbits. This constrains the measure only; a hypothetical packing is not assumed symmetric. Each parent is folded individually, assigned one catalogue core, and then unfolded.

## 6 Reusing certified coverage by recentering cores

The parent-aware theorem above still chooses a core concentric with its parent. That condition is convenient, but it is not part of the counting argument. Only strict containment is required. This observation lets us reuse an already certified core library for nearby parent/container parameters without assuming any continuity of the atomic measure.

Suppose a previously certified core shape  $K_j$  has a certified center domain  $D_j$  and charge at least  $\gamma$  for every translate  $y + K_j$  with  $y \in D_j$ . For a centered parent shape  $P$ , define its *erosion domain*

$$E_j(P) = \{v : v + K_j \subset \text{int } P\}.$$

A parent centered at  $x$  can use this core entry exactly when there is  $y \in D_j$  with  $y - x \in E_j(P)$ , equivalently

$$x \in D_j - E_j(P).$$

For centrally symmetric squares  $E_j(P) = -E_j(P)$ , so this is the Minkowski sum  $D_j + E_j(P)$ . Hence an entire legal parent-center set is covered whenever it lies in the union of these translated erosion domains.

**Proposition 6** (Recentered-core transfer). *Assume a charge  $C$  has budget  $\mathcal{B}$  on pairwise-disjoint cores, and that every translate  $y + K_j$  with  $y \in D_j$  has charge at least  $\gamma$ . If, for every legal parent*

pose  $P + x$ , its center  $x$  lies in

$$\bigcup_j (D_j - E_j(P)),$$

then every parent can choose a strictly interior certified core. If  $\mathcal{B} < n\gamma$ , no packing of  $n$  such parents exists.

*Proof.* Choose an index  $j$  and  $y \in D_j$  with  $y - x \in E_j(P)$ . Then  $y + K_j \subset \text{int}(P + x)$  and has charge at least  $\gamma$ . Apply Lemma 2 to the selected cores.  $\square$

Use one common container-centered coordinate frame for the old and new domains; the new container lies inside the old one. The old measure and its total budget remain available without moving any atoms. For the square domains in the current certificate this condition becomes a single exact scalar inequality. Let the new parent have side  $a$ , angle  $\phi \in [0, \pi/4]$ , and outer side  $H$ , and set

$$f(\phi) = \cos \phi + \sin \phi.$$

Its legal centers form a square of half-side

$$q = \frac{H - af(\phi)}{2}.$$

Suppose old core entry  $j$  has side  $B_j$ , angle  $\theta_j$ , and certified center domain  $[-d_j, d_j]^2$ . Its width across either parent normal is

$$w_j(\phi) = B_j(|\cos(\theta_j - \phi)| + |\sin(\theta_j - \phi)|).$$

**Lemma 7** (Exact square recentering criterion). *Assume  $q \geq 0$  and  $d_j \geq 0$ . Every legal center of the new parent can use entry  $j$  if and only if*

$$\boxed{w_j(\phi) + f(\phi) [H - af(\phi) - 2d_j]_+ < a.} \tag{R}$$

Equivalently,

$$w_j(\phi) < a, \quad H < af(\phi) + 2d_j + \frac{a - w_j(\phi)}{f(\phi)}.$$

*Proof.* Clamp each coordinate of the parent center into  $[-d_j, d_j]$ . Each coordinate moves by at most  $[q - d_j]_+$ . Projection of that displacement on either parent normal is at most  $f(\phi)[q - d_j]_+$ . Adding the core half-width gives sufficiency of (R).

For necessity when  $q > d_j$ , put the parent center at  $(q, q)$ . Every available core center has both coordinates at most  $d_j$ , so its displacement projected on the parent normal  $(\cos \phi, \sin \phi)$  is at least  $f(\phi)(q - d_j)$ . Thus no other recentering can improve the extreme corner. When  $q \leq d_j$ , only  $w_j(\phi) < a$  is needed.  $\square$

This criterion is exact. It reuses mass statements only at core centers that were already certified; no continuity of  $\mu$  under translation is assumed.

## 6.1 A finite scalar sharpening of the base certificate

For old catalogue row  $j$  with parent interval  $I_j$ , define

$$m_j = \min_{u \in I_j} f(u), \quad G_j = \max_{u \in I_j} g(t_j, u), \quad \eta_j = A - B_j G_j.$$

Its old certified center-domain half-side is

$$d_j = \frac{L - A m_j}{2}.$$

Decrease the parent side and outer side to

$$a = A - \varepsilon, \quad H = L - \delta.$$

If  $F_j$  is any upper bound for  $f$  throughout  $I_j$ , then Lemma 7 is implied by the simpler rational condition

$$\boxed{\eta_j > \varepsilon + F_j[\varepsilon F_j - \delta]_+} \quad (\text{S})$$

for every row. Indeed, the bracket bounds the extra center displacement needed to clamp a new legal parent center back into the old certified domain, while  $\eta_j - \varepsilon$  is the remaining strict containment margin.

For the retained 4,391-row catalogue choose

$$\varepsilon = 10^{-11}, \quad \delta = \frac{141}{10^{13}}.$$

Using exact endpoint maxima for  $F_j$  unless the interval crosses the maximum of  $f$ , and the rational upper bound

$$\frac{1414213563}{10^9} > \sqrt{2}$$

otherwise, all 4,391 inequalities (S) hold. The minimum remaining scalar slack is the positive rational

$$\frac{3156837929188515937426608335639966497291}{2009206832324365552240070870206040941943769650000000000} > 0.$$

The replay command evaluates these inequalities exactly in `sharpening.py`, including the minimum slack and the library cap below. No translation coverage calculation is repeated: the old core shapes and old certified center domains are used unchanged.

The old budget inequality

$$17\gamma - M = \frac{4433}{500000000} > 0$$

is likewise unchanged. Therefore the new parents of side  $A - \varepsilon$  do not fit in the new container of side  $L - \delta$ , and scaling gives

$$s(17) > \frac{L - \delta}{A - \varepsilon} = \boxed{\frac{46129999999859}{9997499999900}}.$$

This proves Theorem 1 from the base certificate and finitely many additional rational inequalities.

## 6.2 Why fixed-library recentering is almost exhausted

For an axis-aligned new parent, let

$$W_j = B_j(c(t_j) + s(t_j)).$$

The recentering criterion implies

$$a > W_j, \quad H < 2a + 2d_j - W_j.$$

Hence every argument that pays an axis-aligned parent using one of these unchanged core/domain statements satisfies

$$\frac{H}{a} \leq \max_j \max \left\{ 2, 1 + \frac{2d_j}{W_j} \right\}.$$

For the present core/domain library this cap is

$$\frac{783497431270959447209451212282201903}{169803068915384203003727378193801903} = 4.61415353842273443435 \dots$$

Thus the new rational corollary is already extremely close to the limit of this particular *reuse* strategy. This is not an upper bound on  $s(17)$ , on the underlying measure, or on certificates using new core shapes, new coverage domains, or stronger charge resources.

## 7 The base parent-aware certificate

Set

$$L = \frac{4613}{1000}, \quad A = \frac{3999}{4000}.$$

The exact certificate contains 417 positive  $D_4$  orbit rows, expanding to 3,280 atoms. It partitions the folded orientation range into 4,391 closed parent-angle intervals. The full replay obtains

$$M := \mu(Q_L) = \frac{4249997911}{250000000} = 16.999991644$$

and

$$\gamma = \frac{100000003}{100000000} = 1.000000030.$$

Therefore

$$17\gamma - M = \frac{4433}{500000000} = 0.000008866 > 0.$$

The exact event sweep checks 27,918,671 center slabs. Two accumulation backends—a segment tree and a direct difference-array/prefix engine—agree on every interval minimum. They share the same geometric partition and are therefore not described as independent geometric proofs.

Proposition 3 excludes seventeen side- $A$  parents in  $Q_L$ . Scaling a hypothetical unit-square packing in a container of side  $L/A$  by the factor  $A$  would give precisely such a parent packing. Hence no unit-square packing exists at

$$\frac{L}{A} = \frac{4613/1000}{3999/4000} = \frac{18452}{3999}.$$

Since the minimum defining  $s(17)$  is attained, exclusion at the displayed side gives the strict base theorem

$$s(17) > \frac{18452}{3999} = 4.61415353838459614903 \dots$$

Section 6 then sharpens this base theorem to Theorem 1 without changing the coverage certificate.

The theorem-critical files are under `certificates/lower_bound_4p614153/`. The certificate is stored as raw JSON; the verifier checks its SHA-256 before using it. From the repository root, run

```
python3 certificates/lower_bound_4p614153/verify.py \
--output /tmp/n17 --upstream --direct
```

followed by the package tests. The replay requires Python 3.10+, a C++17 compiler with OpenMP, and Boost headers. Numerical optimization is not in the trusted proof path.

## 8 What R012 unlocked, and where the next limit moved

The improvement from R012 to the present theorem is itself naturally decomposed into three mathematical steps:

| Change                                         | Bound or reported target                                    | Increment over previous row |
|------------------------------------------------|-------------------------------------------------------------|-----------------------------|
| R012 parent-aware catalogue                    | $461300/99999 = 4.613046130461\dots$                        | —                           |
| Adaptive parent-aware cores, same measure      | $922600/199979 = 4.613484415864\dots$                       | $0.000438285402\dots$       |
| Reweight the same original locations           | $230650/49993 = 4.613645910427\dots$                        | $0.000161494564\dots$       |
| Add wall-adjusted support and reweight         | $18452/3999 = 4.614153538385\dots$                          | $0.000507627957\dots$       |
| Analytic recentering of certified core domains | $\frac{46129999999859}{9997499999900}, 4.614153538417\dots$ | $3.20495 \times 10^{-11}$   |

The intermediate adaptive-only and same-support targets are historical search reports; their separate certificates are not retained here. The replay verifies R012, the final base certificate, and the analytic corollary.

This decomposition is useful because the three gains come from different sources. The first removes a verifier overconstraint. The second changes only weights, showing that the parent-aware geometry exposed previously hidden slack in the support. The third changes the support itself, which becomes necessary once an exact low-mass physical parent obstructs the frozen measure.

With Guzhou’s R012 measure held completely fixed, adaptive parent-aware core selection was reported to raise the target to

$$\frac{922600}{199979} = 4.613484415863665\dots$$

An exact legal parent then shows why further angle refinement alone stops helping that frozen measure: at target

$$\frac{461300}{99989} = 4.613507485823440\dots$$

the entire parent captures only 0.998982 mass, below the required common charge ratio. No inner core of that parent can capture more mass than the parent itself. This is a limitation of that fixed measure under the common charge rule, not an upper bound on  $s(17)$ .

The final improvement therefore changes the measure. Reweighting the inherited support first passes the fixed-measure obstruction; mixing retained sites with wall-adjusted copies then reaches  $A = 3999/4000$  and the headline theorem. This separates two kinds of progress that had previously been conflated: improving the finite verifier, and improving the underlying measure.

The remaining gap to Bidwell is still substantial. Other work in this repository studies local optimality and compatibility-aware strengthenings, but no global proof of Bidwell optimality is claimed here.

## 9 Compatibility information as spatial charges

The point-measure method prices each core through additive spatial mass. That relaxation can miss information about which individually admissible square poses can coexist. We now describe a budgeted charge that converts finite overlap incompatibilities back into the same spatial containment language used by the event sweep.

### 9.1 Intersection-trigger atoms

Let  $T_1, \dots, T_m$  be nonempty finite sets of spatial points satisfying

$$T_i \cap T_j \neq \emptyset \quad (i \neq j).$$

Define

$$C_T(K) = \mathbf{1}\{\exists i : T_i \subseteq K\}.$$

**Proposition 8** (Budget-one intersection trigger). *For every pairwise-disjoint family of closed cores  $K_1, \dots, K_r$ ,*

$$\sum_{\ell=1}^r C_T(K_\ell) \leq 1.$$

*Proof.* If two disjoint cores were charged, they would contain triggers  $T_i$  and  $T_j$ . If  $i = j$ , both cores contain the same nonempty set. If  $i \neq j$ , both contain a point in  $T_i \cap T_j$ . Either conclusion contradicts disjointness.  $\square$

Levy’s two-of-three threshold atom is a special case, obtained by taking the three two-element subsets of a three-point set as triggers. More generally, if the triggers are not pairwise intersecting, their matching number—the largest number of pairwise-disjoint triggers—is a valid global budget. Thus Lemma 2 simultaneously contains point measures, threshold atoms, and these compatibility charges.

### 9.2 Compiling an overlap clique into one spatial resource

Suppose  $m \geq 2$  square poses  $V_1, \dots, V_m$  have pairwise-overlapping interiors. Choose, for every pair, a common interior point

$$p_{ij} = p_{ji} \in \text{int}(V_i) \cap \text{int}(V_j),$$

and define

$$T_i = \{p_{ij} : j \neq i\}.$$

Then  $V_i$  contains  $T_i$ , while  $p_{ij} \in T_i \cap T_j$ . Proposition 8 therefore gives one budget-one resource activating on the whole clique. Strict overlap provides open pose neighborhoods of activation, so this construction is not tied to isolated exact poses.

The retained example in `certificates/intersection_trigger_29/` contains 29 rational unit-square poses, 406 distinct pair witnesses, and 29 triggers of size 28. The minimum witness margin inside its required source unit square is

$$\frac{314856211897415969021}{287142759881684900000000} > \frac{1}{1000}.$$

Thus every trigger remains inside its corresponding concentric closed side-499/500 core  $K_i$ , and  $C_T(K_i) = 1$  for all 29 cores.

The supplied nonnegative weights  $\lambda_i$  satisfy

$$S := \sum_i \lambda_i = \frac{81407286808}{79999999999}, \quad \max_x \sum_i \lambda_i \mathbf{1}_{\text{int } V_i}(x) = m := \frac{59431493389}{79999999999}.$$

The checker verifies this maximum directly on the 29 poses. It enumerates all vertex and edge-intersection abscissae, then all vertical interval cells in each of the resulting 1,475 open slabs. The covering set is constant in each cell. A point on an arrangement boundary cannot have a greater open-square load: every open square containing that point also contains a neighbourhood of it. A rational maximum witness is directly recounted.

Since  $K_i \subset \text{int } V_i$ , the same pointwise upper bound holds for the closed cores, including their boundaries. Therefore any nonnegative measure  $\nu$  dominating the trigger charge on these cores satisfies

$$S \leq \sum_i \lambda_i \nu(K_i) = \int \sum_i \lambda_i \mathbf{1}_{K_i} d\nu \leq m \nu(\mathbb{R}^2).$$

Its total mass is consequently at least

$$\nu(\mathbb{R}^2) \geq \frac{S}{m} = \frac{81407286808}{59431493389} = 1.3697668048682659 \dots > 1,$$

whereas the trigger resource has budget one. This proves a strict separation on the supplied finite family. It uses only these 29 poses, without assuming a load theorem for a larger source family, and does not change the numerical bound on  $s(17)$ .

The source expansion also verifies 71 recorded unit-square activations with weight sum  $99656162199/79999999999$ , of which 68 survive shrinking to side 499/500, with weight sum  $96836866527/79999999999$ . An independent 8,405-slab arrangement check on those 71 poses gives maximum pointwise load  $73959042636/79999999999 < 1$ . This verifies the earlier 68-core separation of  $1.2104608316026308 \dots$  without assuming the full source-family load claim; the normalized 29-pose argument above gives the stronger separation.

### 9.3 The continuous geometry remains two-dimensional

At a fixed core direction, rotate every point  $p \in T_i$  to coordinates  $(u_p, v_p)$ . A side- $B$  core centered at  $(u, v)$  contains the entire trigger  $T_i$  exactly when its center lies in the rectangle

$$R_i = \left[ \max_{p \in T_i} u_p - \frac{B}{2}, \min_{p \in T_i} u_p + \frac{B}{2} \right] \times \left[ \max_{p \in T_i} v_p - \frac{B}{2}, \min_{p \in T_i} v_p + \frac{B}{2} \right],$$

with a reversed interval interpreted as empty. Hence the charge is the indicator of

$$R_1 \cup \dots \cup R_m.$$

For  $m$  triggers this is a union of at most  $m$  rectangles at each direction. Standard vertical-slab decomposition gives polynomial-size exact event geometry. The pieces remain one resource with one budget; decomposing the union must not assign independent budgets to the pieces.

For a general finite overlap graph  $G$ , choose one shared interior witness per certified overlap edge and let each vertex trigger contain its incident edge witnesses. Give each isolated vertex a private interior witness as well, so every trigger is nonempty. Charged disjoint cores select distinct, nonadjacent vertices of  $G$ , so any proved bound  $\alpha(G) \leq b$  yields budget  $b$ . Cliques give  $b = 1$ , while odd cycles and more complicated graphs can encode compatibility information not expressible by clique cuts alone.

The current numerical lower bound does *not* use intersection-trigger atoms. This section supplies a general mathematical framework for future certificates: the parent-aware and recentering theorems apply unchanged to a total charge

$$C(K) = \mu(K) + \sum_r w_r C_{T(r)}(K)$$

with nonnegative weights  $w_r$  and budget

$$\mu(Q_L) + \sum_r w_r b_r.$$

A future improvement requires proving the corresponding universal core-charge lower bound.

## 10 Contributors, attribution, and verification status

The present theorem and its analytic sharpening have one author, but they sit in an unusually collaborative and rapidly evolving computational lineage. Direct mathematical and software dependencies are credited in the certificate’s `ATTRIBUTION.md`; the repository’s `CONTRIBUTORS.md` gives a broader chronological account with links to the public repositories, papers, and audit sources.

In particular, the current parent-aware implementation uses the conceptual lineage of Levy’s weighted/event-sweep work and Guzhou0806’s concrete R012 catalogue, while its spatial support descends from this repository’s earlier measure. The exact R012 source data used for replay are pinned to commit `931a0dfd64302e277057006e99388fe5c00b7f53`. The retained copy of the R012 Python sweep preserves Joshua Levy’s MIT notice. Fort, Burns, Massaccesi, anabologyco-maker, MacIver, Green, Stromquist, Nagamochi, Bidwell, and others are cited for historical or parallel contributions as specified in `CONTRIBUTORS.md`; citation does not imply endorsement or coauthorship.

The computation and exposition were developed with substantial AI assistance under human direction. The mathematical claim rests on explicit rational data, the stated reduction, and executable exact verification. This is a computer-assisted working preprint. Independent external review and formal proof-assistant verification of the complete parent-aware argument remain open.

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